

→ PROPA

ACTIVIDAD COMPLEMENTARIA.

APARADO A.

BINARIO A DECIMAL.

a) $101010_2 \rightarrow ()_{10}$

$$\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{array} = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$
$$= 32 + 0 + 8 + 0 + 2 + 0$$

$$101010 = 42_{10}$$

b) $111001_2 \rightarrow ()_{10}$

$$\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$
$$= 32 + 16 + 8 + 0 + 0 + 1$$

$$111001 = 57_{10}$$

c) $10011_2 \rightarrow ()_{10}$

$$\begin{array}{ccccc} 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{array} = 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
$$= 16 + 0 + 0 + 2 + 1$$

$$10011_2 = 19_{10}$$

d) $1101,01_2 \rightarrow ()_{10}$

$$\begin{array}{cccccc} 3 & 2 & 1 & 0 & -1 & -2 \\ 1 & 1 & 0 & 1 & 0 & 1 \end{array}$$
$$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0, 0 \times 2^{-1} + 1 \times 2^{-2}$$
$$8 + 4 + 0 + 1, 0 + \frac{1}{4}$$
$$(13,25)_{10}$$

$$e) 10110,1_2$$

$$\begin{array}{cccccc} 4 & 3 & 2 & 1 & 0 & -1 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ \times 2^4 & 0 \times 2^3 & 1 \times 2^2 & 1 \times 2^1 & 0 \times 2^0 & 1 \times 2^{-1} \\ 16 & + 0 & + 4 & + 2 & + 0 & + \frac{1}{2} \\ & & & & & (22,5)_{10} \end{array}$$

$$f) 10001,11_2$$

$$\begin{array}{cccccc} 4 & 3 & 2 & 1 & 0 & -1 & -2 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ \times 2^4 & 0 \times 2^3 & 0 \times 2^2 & 0 \times 2^1 & 1 \times 2^0 & 1 \times 2^{-1} & 1 \times 2^{-2} \\ 16 & + 0 & + 0 & + 0 & + 1 & + 0,5 & + 0,25 \\ & & & & & & (17,75)_{10} \end{array}$$

PARTE B DECIMAL A BINARIO

$$a) 12_{10} \rightarrow ()_2$$

$$\begin{array}{r} 12 \div 2 \\ 0 \quad 6 \div 2 \\ 0 \quad 3 \div 2 \\ 0 \quad 1 \end{array}$$

$$12_{10} = 1100_2$$

$$b) 25_{10} \rightarrow ()_2$$

$$\begin{array}{r} 25 \div 2 \\ 1 \quad 12 \div 2 \\ 0 \quad 6 \div 2 \\ 0 \quad 3 \div 2 \\ 1 \quad 1 \end{array}$$

$$25_{10} = 11001_2$$

$$c) 45_{10} \rightarrow ()_2$$

$$\begin{array}{r} 45 \div 2 \\ 1 \quad 22 \div 2 \\ 0 \quad 11 \div 2 \\ 1 \quad 5 \div 2 \\ 1 \quad 2 \div 2 \\ 0 \quad 1 \end{array}$$

$$45_{10} = 101101_2$$

$$d) 64_{10} \rightarrow ()_2$$

$$\begin{array}{r} 64 \div 2 \\ 0 \quad 32 \div 2 \\ 0 \quad 16 \div 2 \\ 0 \quad 8 \div 2 \\ 0 \quad 4 \div 2 \\ 0 \quad 2 \div 2 \\ 0 \quad 1 \end{array}$$

$$64_{10} = 1000000_2$$

$$e) 22,5_{10} \rightarrow ()_2$$

$$\begin{array}{r} 22 \div 2 \\ 0 \quad 11 \div 2 \\ 1 \quad 5 \div 2 \\ 1 \quad 2 \div 2 \\ 0 \quad 1 \end{array}$$

$$22,5_{10} = 10110,1_2$$

$$0,5 \times 2 = 1,00$$

$$f) (10,25)_{10} \rightarrow ()_2$$

$$\begin{array}{r} 10 \div 2 \\ 0 \quad 5 \div 2 \\ 1 \quad 2 \div 2 \\ 0 \quad 1 \end{array}$$

$$(10,25)_{10} = 1010,01_2$$

$$0,25 \times 2 = 0,5 \rightarrow 0$$

$$0,5 \times 2 = 1,0 \rightarrow 1$$

$$g) (18,75)_{10} \rightarrow ()_2$$

$$\begin{array}{r} 18 \div 2 \\ 0 \quad 9 \div 2 \\ 1 \quad 4 \div 2 \\ 0 \quad 2 \div 2 \\ 0 \quad 1 \end{array}$$

$$(100101,11)_2$$

$$0,75 \times 2 = 1,50 \rightarrow 1$$

$$0,50 \times 2 = 1,00 \rightarrow 1$$

→ ACTIVIDAD COMPLEMENTARIA.

EJ. 1. REDONDEO VS TRUNCAMIENTO

SEA EL NUMERO $x = 3,78654$

① Aprox a 3 cifras decimales por:

a) REDONDEO

$$\begin{array}{r} 3,78654 \\ 3,787 \end{array}$$

b) TRUNCAMIENTO

$$3,786 //$$

② CALCULAR:

ERROR ABSOLUTO

$$E_a = |x - \bar{x}| \quad \begin{array}{l} x = \text{E. REAL} \\ \bar{x} = \text{E. Aprox} \end{array}$$

REDONDEO

$$E_a = |3,78654 - 3,787|$$

$$E_a = 0,00046$$

TRUNCAMIENTO

$$E_a = |3,78654 - 3,786|$$

$$E_a = 0,00054$$

ERROR RELATIVO

$$E_r = \frac{|x - \bar{x}|}{|x|}$$

$$E_r = \frac{0,00046}{3,78654}$$

$$E_r = 0,0001215$$

$$E_r = \frac{0,00054}{3,78654}$$

$$E_r = 0,0001426$$

EJ 2. OPERACIONES CON APROXIMACIONES

SEA. $a = 5,347$ y $b = 2,918$

• APROXIME AMBOS A 2 DECIMALES USANDO

REDONDEO

$$\begin{array}{r} a) \rightarrow 5,347 \\ 5,35 \end{array}$$

$$\begin{array}{r} b) \rightarrow 2,918 \\ 2,92 \end{array}$$

$$\text{suma } (5,35 + 2,92)$$

$$8,27$$

TRUNCAMIENTO

$$a) \rightarrow 5,34$$

$$b) \rightarrow 2,91$$

$$\text{suma } (5,34 + 2,91)$$

$$8,25$$

② Suma REAL

$$\begin{array}{r} 5,347 + 2,918 \\ 8,265 \end{array}$$

E_a x TRUNCAMIENTO

$$\begin{aligned} E_{at} &= |8,265 - 8,25| \\ &= 0,015 \end{aligned}$$

E_a x REDONDEO

$$\begin{aligned} E_{ar} &= |8,265 - 8,27| \\ &= 0,005 \end{aligned}$$

③ Metodo + preciso $0,005 < 0,015$ ES POR REDONDEO